Cloud Coputing
Lab-3

Name: PES1UG23MA343
Section: F
SRN:PES1UG23AM343

```
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent
it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
Run 'docker --help' for more information
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker images
                                                    i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest             05813aedc15f       25.9kB         9.52kB    U
python:3.10-slim               f5d029fe3914        185MB         47.1MB
spam-web-app-pes1ug23am343:latest
                               491f1199284c        541MB          125MB
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
                               491f1199284c        541MB          125MB
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker pull python:3.10-slim
3.10-slim: Pulling from library/python
63ebb7511fc1: Pull complete
0c8d55a45c0d: Pull complete
d0d1977db485: Pull complete
690eaffcf0e9: Pull complete
13a7f29dd8f5: Download complete
a4e5ac615dd3: Download complete
Digest: sha256:218027ad8e5a6fb68245ccffcb8bcfa98d9891039bfedd53b1b4666dc48a4
bb1
Status: Downloaded newer image for python:3.10-slim
docker.io/library/python:3.10-slim
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker images
                                                    i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest             05813aedc15f       25.9kB         9.52kB    U
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docker pull python:3.10-slim
3.10-slim: Pulling from library/python
63ebb7511fc1: Pull complete
0c8d55a45c0d: Pull complete
d0d1977db485: Pull complete
690eaffcf0e9: Pull complete
13a7f29dd8f5: Download complete
a4e5ac615dd3: Download complete
Digest: sha256:218027ad8e5a6fb68245ccffcb8bcfa98d9891039bfedd53b1b4666dc48a4
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spam-web-app-pes1ug23am343:latest
                               491f1199284c        541MB          125MB
```

```
Run 'docker buildx build --help' for more information
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker build -t spam-web-app-pes1ug23am343 .
[+] Building 54.6s (12/12) FINISHED                          docker:default
 => [internal] load build definition from Dockerfile                    0.0s
 => => transferring dockerfile: 281B                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim     0.1s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 2B                                         0.0s
 => [1/7] FROM docker.io/library/python:3.10-slim@sha256:218027ad8e5a   0.2s
 => => resolve docker.io/library/python:3.10-slim@sha256:218027ad8e5a   0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 184B                                       0.0s
 => [2/7] WORKDIR /app                                                  0.1s
 => [3/7] COPY app.py ./                                                0.1s
 => [4/7] COPY spam_classifier_model.pkl ./                             0.1s
 => [5/7] COPY tfidf_vectorizer.pkl ./                                  0.1s
 => [6/7] COPY templates ./templates                                    0.1s
 => [7/7] RUN pip install --no-cache-dir flask scikit-learn joblib     25.7s
 => exporting to image                                                 27.4s
 => => exporting layers                                                14.1s
 => => exporting manifest sha256:27587433dbdf22f00e08107d057a0f23accd   0.4s
 => => exporting config sha256:412f4e060b97c5058d546462b4335bdaa991f1   1.7s
 => => exporting attestation manifest sha256:906c1d7ad237332e50b45d53   2.5s
 => => exporting manifest list sha256:0c176effa111cd65199b6b678af35d4   1.5s
 => => naming to docker.io/library/spam-web-app-pes1ug23am343:latest    0.6s
 => => unpacking to docker.io/library/spam-web-app-pes1ug23am343:late   4.2s
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker run -d -p 5000:5000 spam-web-app-pes1ug23am343
3f2ce948ab8ffe207ab03cf117a477871739886826d4d06dc25954adff55af62
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker ps
CONTAINER ID   IMAGE                          COMMAND           CREATED
  STATUS         PORTS                                            NAMES
3f2ce948ab8f   spam-web-app-pes1ug23am343   "python app.py"   7 seconds ago
  Up 6 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   stoic_archime
des
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
des
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login
-u <username>'


Your one-time device confirmation code is: VRLX-XKPN
Press ENTER to open your browser or submit your device code here: https://lo
gin.docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/home/vardha/.docker/co
nfig.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker tag spam-web-app-pes1ug23am343 vardhakathuria/spam-web-app-pes1ug23am
343:v1
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker images
                                                  i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest             05813aedc15f       25.9kB         9.52kB    U
python:3.10-slim               218027ad8e5a        185MB         47.1MB
spam-web-app-pes1ug23am343:latest
                               0c176effa111        541MB          125MB    U
vardhakathuria/spam-web-app-pes1ug23am343:v1
                               0c176effa111        541MB          125MB    U
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

PES1UG23AM343

Spam Detection using ML (Dockerized)

hello

Predict

Prediction: NOT SPAM

PES1UG23AM343

Spam Detection using ML (Dockerized)

Enter message

Predict

Prediction: NOT SPAM

```
tag does not exist: vardhakathuria/spam-web-app-pes1ug23am34:v1
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker push vardhakathuria/spam-web-app-pes1ug23am343:v1
The push refers to repository [docker.io/vardhakathuria/spam-web-app-pes1ug2
3am343]
690eaffcf0e9: Mounted from library/python
306ea65694ec: Pushed
00e279e7e6ef: Pushed
365942a0197e: Pushed
f8ed1aec6232: Pushed
019c7c71d7e0: Pushed
63ebb7511fc1: Mounted from library/python
0c8d55a45c0d: Mounted from library/python
2b6dde27c8c1: Pushed
d0d1977db485: Mounted from library/python
666efda71fdd: Pushed
v1: digest: sha256:0c176effa111cd65199b6b678af35d49ddf2ab1df9940eb7bfa85c4d6
ce0ca0b size: 856
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

Images Give feedback

Local My Hub

415.72 MB / 185.21 MB in use 4 images

Last refresh: 12 hours ago

Search

Name	Tag	Image ID	Created	Size	Actions
hello-world	latest	05813aedc15f	6 months ago	25.9 KB	
<none>	<none>	f5d029fe3914	23 days ago	185.06 MB	
python	3.10-slim	218027ad8e5a	8 hours ago	185.07 MB	
spam-web-app-pes1ug2:	latest	0c176effa111	7 minutes ago	540.87 MB	
vardhakathuria/spam-we	v1	0c176effa111	7 minutes ago	540.87 MB	

```
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker ps
CONTAINER ID   IMAGE                          COMMAND           CREATED
  STATUS         PORTS                                            NAMES
3f2ce948ab8f   spam-web-app-pes1ug23am343   "python app.py"   6 minutes ago
  Up 6 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   stoic_archime
des
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker stop 3f2ce948ab8f
3f2ce948ab8f
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker ps
CONTAINER ID   IMAGE                          COMMAND           CREATED
  STATUS         PORTS                                            NAMES
3f2ce948ab8f   spam-web-app-pes1ug23am343   "python app.py"   6 minutes ago
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vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker stop 3f2ce948ab8f
3f2ce948ab8f
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker rmi spam-web-app-pes1ug23am343
Untagged: spam-web-app-pes1ug23am343:latest
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker images
                                                    i Info →   U  In Use
IMAGE                        ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest           05813aedc15f       25.9kB         9.52kB    U
python:3.10-slim             218027ad8e5a        185MB         47.1MB
vardhakathuria/spam-web-app-pes1ug23am343:v1
                             0c176effa111        541MB          125MB    U
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker rmi vardhakathuria/spam-web-app-pes1ug23am343:v1
Error response from daemon: conflict: unable to delete vardhakathuria/spam-w
eb-app-pes1ug23am343:v1 (must be forced) - container 3f2ce948ab8f is using i
ts referenced image 0c176effa111
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```

```
                                                    i Info →   U  In Use
IMAGE   ID             DISK USAGE   CONTENT SIZE   EXTRA

No images found matching "prune": did you mean "docker image prune"?
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
docker pull vardhakathuria/spam-web-app-pes1ug23am343:v1
v1: Pulling from vardhakathuria/spam-web-app-pes1ug23am343
Digest: sha256:0c176effa111cd65199b6b678af35d49ddf2ab1df9940eb7bfa85c4d6ce0c
a0b
Status: Image is up to date for vardhakathuria/spam-web-app-pes1ug23am343:v1
docker.io/vardhakathuria/spam-web-app-pes1ug23am343:v1
vardha@DESKTOP-2BHUB51:/mnt/c/Users/acer/Downloads/ml-docker-PES1UG23AM343$
```